Modular Forms and Quantum Black Holes

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CERTIFICATE

This is to certify that the work contained in this project report entitled **"Modular Forms and Quantum Black Holes"** submitted by **Tharun Naagesh** (**Roll No: IMS22252**) to Indian Institute of Science Education and Research, Thiruvananthapuram towards the partial requirement of **Minor Degree** in **Physics** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

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Contents

Chapter 1

Introduction to Modular Forms

Abstract

This report studies the theory of modular forms and Jacobi forms, and their role in counting problems arising in the context of quantum black holes [DMZ12]. We begin with a brief review of Fourier series [Sch12] and classical modular forms [DS05], before developing the theory of modular forms for congruence subgroups and Jacobi forms. We then discuss their applications to string theory, including a correspondence between type IIA string theory on $K3 \times T^2$ and the heterotic string on $T^4 \times T^2$ at the level of partition functions. We also give a physical interpretation of the Fourier coefficients of $1/\Delta$ (see Appendix C.4.2).

1.1 Fourier Series

Let $\mathcal{S}$ denote the space of complex-valued, $\mathbb{Z}$-periodic functions on $\mathbb{R}$. Let the inner product be the standard Hermitian inner product [Sch12, Section 1.2]. For $f \in \mathcal{S}$, define the Fourier coefficients

$$c_n(f) := \int_0^1 f(x) e^{-2\pi i n x}\, dx, \quad n \in \mathbb{Z}.$$

The Fourier series of f is then

$$f(x) \sim \sum_{n \in \mathbb{Z}} c_n(f)\, e^{2\pi i n x}.$$

1.2 Modular Forms

1.2.1 Basic Definitions and Results

Definition 1.2.1 (Basic objects)**.**

$$\begin{aligned} \mathbb{H} &= \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\} \\ SL_2(\mathbb{R}) &= \{A \in M_2(\mathbb{R}) : \det A = 1\} \\ GL_2(\mathbb{R}) &= \{A \in M_2(\mathbb{R}) : \det A \neq 0\} \\ GL_2^+(\mathbb{R}) &= \{A \in M_2(\mathbb{R}) : \det A > 0\}. \end{aligned}$$

Definition 1.2.2 (Modular form for $SL_2(\mathbb{Z})$)**.** Let $k \in \mathbb{Z}$. A function $f : \mathbb{H} \to \mathbb{C}$ is a modular form of weight k if:

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \quad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}),\ z \in \mathbb{H},$$

and f is holomorphic on $\mathbb{H}$ and bounded as $\mathrm{Im}(z) \to \infty$.

Corollary 1.2.3. For $f \in M_k(SL_2(\mathbb{Z}))$,

$$f(z+1) = f(z), \quad f(-1/z) = z^k f(z), \quad f(z) = (-1)^k f(z).$$

In particular, if k is odd then $f \equiv 0$.

Proof. Since $T, S, -I \in SL_2(\mathbb{Z})$, the result follows.

□

Fact 1.2.4. $SL(2,\mathbb{Z})$ is generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. [Con16, Theorem 3.3]

Definition 1.2.5 (Weakly holomorphic modular forms)**.** A function $f : \mathbb{H} \to \mathbb{C}$ is a weakly holomorphic modular form of weight k if

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau)$$

for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, and f is holomorphic on $\mathbb{H}$ with a Fourier expansion

$$f(\tau) = \sum_{n \geq -N} a_n q^n, \quad q = e^{2\pi i \tau},$$

for some $N \geq 0$.

Definition 1.2.6. Let M_k, S_k, and $M_k^!$ denote respectively the spaces of modular forms, cusp forms, and weakly holomorphic modular forms of weight k (all over $\mathbb{C}$). Thus,

$$S_k \subset M_k \subset M_k^!.$$

Remark 1.2.7. When necessary, we specify the group, e.g. $M_k(SL_2(\mathbb{Z}))$ or $S_k(SL_2(\mathbb{Z}))$.

Growth of Fourier coefficients ($SL_2(\mathbb{Z})$)

For $f(\tau) = \sum a_n q^n$, we have:

$$\begin{aligned} f \in S_k &\implies a_n = O(n^{k/2}), \\ f \in M_k &\implies a_n = O(n^{k-1}), \\ f \in M_k^! &\implies a_n = O(e^{C\sqrt{n}}) \end{aligned}$$

[DMZ12] [Con16, Section 5].

Definition 1.2.8. (Eisenstein Series E_k)

$$E_k(z) := \sum_{(m,n)\in\mathbb{Z}^2, (m,n)\neq 0} \frac{1}{(mz+n)^k}$$

Remark 1.2.9. $E_k(z)$ is a modular form of weight k (for $SL_2(\mathbb{Z})$) for all $k \geq 4$. They are very important examples of modular forms for they, in some sense, form the "building blocks" to construct other modular forms.

Definition 1.2.10. (Discriminant function Δ)

$$\Delta(z) := q\Pi_1^\infty(1-q^n)^{24}$$

(Where $q = e^{2\pi i z}$). This function is a *cusp form* of weight 12.

Definition 1.2.11. (Dedekind Eta Function η)

$$\eta(z) := q^{1/24}\Pi_1^\infty(1-q^n)$$

(Where $q = e^{2\pi i z}$). This function is a *weakly holomorphic form* of weight 1/2. For more discussion, see [CP+22, Section 3.9].

1.2.2 Structure of $M_k(SL_2(\mathbb{Z}))$ and $S_k(SL_2(\mathbb{Z}))$

We present, without proof, one of the cornerstone theorems in Modular Forms theory:

Theorem 1.2.12. Structure Theorem

1. $M_k(SL_2(\mathbb{Z}))$ is generated by the monomials $E_4^\alpha E_6^\beta$ such that $4\alpha+6\beta=k$. In other words, as a vector space we see that
$$M_k(SL_2(\mathbb{Z})) = \langle\{E_4^\alpha E_6^\beta : 4\alpha+6\beta=k \text{ (in integers)}\}\rangle$$
A trivial corollary of this is that, $M_k(SL_2(\mathbb{Z}))$ is a finite dimensioned vector space.

2. $$M_*(SL_2(\mathbb{Z})) = \oplus_k M_k(SL_2(\mathbb{Z}))$$
forms a *Graded Algebra*- A direct sum of submodules that forms an algebra. Moreover,
$$M_*(SL_2(\mathbb{Z})) = \mathbb{C}[E_4, E_6]$$
This says that, in principle, the space of all modular forms for $SL_2(\mathbb{Z})$ acts as the set of all 2 variable polynomials over $\mathbb{C}$ (E_4 and E_6 are algebraically independent).

3. $$M_k(SL_2(\mathbb{Z})) = \mathbb{C}\cdot E_k + S_k$$
This result says that every modular form is some scalar times the Eisenstein, with the "error term" being a cusp form (or the other way around, where every modular form is a cusp form, with the error term being a factor of the Eisenstein)

4. $$S_k = \Delta\cdot M_{k-12}$$

Proof. [Con16, Section 5], and [Sch12, Theorem 3.9](The Riemann-Roch Theorem) discuss this in greater detail.

□

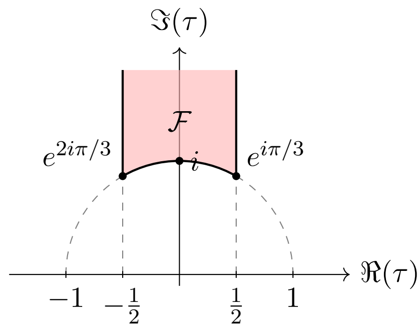
Remark 1.2.13. With the results 3 and 4 of 1.2.12, we see that $M_k = \mathbb{C}E_k + \Delta M_{k-12} = \mathbb{C}E_k + \Delta(\mathbb{C}E_{k-12} + \Delta M_{k-24}) = \mathbb{C}E_k + \Delta(\mathbb{C}E_{k-12} + \Delta(\mathbb{C}E_{k-24} + \Delta M_{k-36})) = \cdots$ from which we get that any modular form f of weight k can be written as
$$f = \sum_{n=0}^{[k/12]} \alpha_n E_{k-12n}\Delta^n$$

1.2.3 Fundamental Domain and the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$

The defining action of a $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL_2(Z)$ on a vector $z\in\mathbb{H}$ is given by:
$$z \mapsto \gamma z = \frac{az+b}{cz+d}$$

Definition 1.2.14 (Fundamental domain for $\Gamma \leq SL_2(\mathbb{Z})$)**.** A subset $\mathcal{F}\subset\mathbb{H}$ is a fundamental domain for Γ [Pra10] if $\mathbb{H} = \bigcup_{\gamma\in\Gamma}\gamma\mathcal{F}$, $(\gamma_1\mathcal{F})^\circ \cap (\gamma_2\mathcal{F})^\circ = \varnothing$ for $\gamma_1\neq\gamma_2$, and $\mathcal{F} = \overline{\mathcal{F}^\circ}$.

Example 1.2.15. A fundamental Domain for $SL_2(\mathbb{Z})$ is
$$\mathcal{F}_{SL_2(\mathbb{Z})} := \{x\in\mathbb{H} : |\operatorname{Re}(x)| \leq \frac{1}{2} \text{ and } |x|>1\}$$



1.2.4 Shimura slash operator

For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2^+(\mathbb{Q})$,

$$(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz+d)^{-k} f\left(\frac{az+b}{cz+d}\right),$$

with $(f|_k\gamma_1)|_k\gamma_2 = f|_k(\gamma_1\gamma_2)$ [Pra10, p. 9].

1.2.5 Congruence Subgroups of $SL_2(\mathbb{Z})$ and Modular Forms for Congruence Subgroups

Definition 1.2.16. ($\Gamma(N)$) For $N \in \mathbb{N}$, we define the principal congruence subgroup of level N as

$$\Gamma(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \textbf{ mod } N \right\}$$

It is not difficult to show that $\Gamma(N)$ is a normal subgroup of $SL_2(\mathbb{Z})$, and that the quotient group $SL_2(\mathbb{Z})/\Gamma(N)$ is isomorphic to $SL_2(\mathbb{Z}/N\mathbb{Z})$ (Hence, $\Gamma(N)$ is a finite index subgroup of $SL_2(\mathbb{Z})$).

Definition 1.2.17. ($\Gamma_0(N)$) For $N \in \mathbb{N}$, we define the congruence subgroup of level N as

$$\Gamma_0(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : c \equiv 0 \textbf{ mod } N \right\}$$

Definition 1.2.18. ($\Gamma_1(N)$) For $N \in \mathbb{N}$, we define the congruence subgroup of level N as

$$\Gamma_1(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) : a \equiv d \equiv 1 \textbf{ mod } N \textbf{ and } c \equiv 0 \textbf{ mod } N \right\}$$

Remark 1.2.19. $\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset SL_2(\mathbb{Z})$

Definition 1.2.20. (Modular Forms for Γ (where Γ is a congruence subgroup of $SL_2(\mathbb{Z})$)) Let Γ be a congruence subgroup of $SL_2(\mathbb{Z})$ and $k \in \mathbb{Z}$. A function $f : \mathbb{H} \to \mathbb{C}$ is a modular form of weight k for Γ if:

1. $f(z)|_k[\gamma] = f(z)$ for all $\gamma \in \Gamma$
2. $\forall\gamma \in SL_2(\mathbb{Z})$, $f(z)|_k[\gamma] = \sum_{n\geq 0} a_n q^{n/N}$

Remark 1.2.21. The reason why we need the full $SL_2(\mathbb{Z})$ in the second condition is that, we need to ensure that f is holomorphic at all the cusps of Γ. For more on this, refer to Appendix A and [Mas16, Section 2.2].

We now briefly introduce Jacobi Forms:

Definition 1.2.22. (Jacobi Forms) Let $k, m \in \mathbb{Z}$. A holomorphic function $\phi : \mathbb{H} \times \mathbb{C} \to \mathbb{C}$ is a Jacobi form of weight k and index m if:

- $\phi(\frac{a\tau+b}{c\tau+d}, \frac{z}{c\tau+d}) = (c\tau+d)^k e^{2\pi i m \frac{cz^2}{c\tau+d}} \phi(\tau, z)$ for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$
- $\phi(\tau, z + \lambda\tau + \mu) = e^{-2\pi i m(\lambda^2\tau + 2\lambda z)}\phi(\tau, z)$ for all $\lambda, \mu \in \mathbb{Z}$
- $\phi(\tau, z) = \sum_{n,r} c(n, r) q^n \zeta^r$ where $q = e^{2\pi i\tau}$ and $\zeta = e^{2\pi i z}$, and $c(n, r) = 0$ unless $4nm \geq r^2$ (This condition is called the "holomorphicity condition" and ensures that ϕ is holomorphic at all the cusps of the Jacobi group).

Chapter 2

Quantum Black Holes and Modular Forms

Classically, a black hole is characterised entirely by its mass, charge, and angular momentum (the "no-hair theorem"). In particular, it has no internal structure.

Bekenstein and Hawking showed in the 1970s that a black hole carries an entropy proportional to the area of its event horizon:

$$S_{\mathrm{BH}} = \frac{A}{4G_N} \tag{2.0.1}$$

where A is the horizon area and G_N is Newton's constant. This is a macroscopic, thermodynamic entropy. By the Boltzmann relation $S = \log \Omega$, this entropy should equal the logarithm of the number of microscopic quantum states Ω of the black hole.

We consider a particular class of black holes in string theory, known as BPS black holes. The partition function $Z(\tau)$ that counts the degeneracies of these BPS states turns out to be a modular form, as we shall see.

2.1 Partition Functions and BPS State Counting

We now explain how modular forms arise naturally in the counting of BPS states, following the discussion in [DMZ12]. A lot more is discussed in appendix C.

The Fock Space $\mathcal{H}$ is built from bosonic oscillators in 24 transverse spatial directions. For each direction $i \in \{1, \ldots, 24\}$ and each mode number $n \in \{1, 2, 3, \ldots\}$, we have a pair of creation/annihilation operators $a_{in}^\dagger$, a_{in} satisfying the canonical commutation relations:

$$[a_{in},\, a_{jm}^\dagger] = \delta_{ij}\, \delta_{n+m,0} \qquad i, j \in \{1, \ldots, 24\}, \quad n, m \in \{1, 2, \ldots\} \tag{2.1.1}$$

All other commutators vanish: $[a_{in}, a_{jm}] = 0$ and $[a_{in}^\dagger, a_{jm}^\dagger] = 0$.

The **Hamiltonian** is:

$$H = \sum_{i=1}^{24} \sum_{n=1}^{\infty} n\, a_{in}^\dagger a_{in} - 1 \tag{2.1.2}$$

The partition function is given by (Appendix C.2.2)

$$Z(\tau) = Tr_{\mathcal{H}} \left(q^H \right) = q^{-1} \prod_{n=1}^{\infty} \frac{1}{(1 - q^n)^{24}} = \frac{1}{\Delta(\tau)} \tag{2.1.3}$$

Where

$$\Delta(\tau) = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

This gives us

$$\boxed{Z(\tau) = \frac{1}{\Delta(\tau)}} \tag{2.1.4}$$

From Appendix C.2.3, we see that $Z(\tau)$ is a weakly holomorphic modular form of weight -12. It has a simple pole at the cusp $\tau = i\infty$, and is holomorphic everywhere else on $\mathbb{H}$. It therefore has a q-expansion of the form

$$Z(\tau) = \sum_{n=-1}^{\infty} d(n) q^n \tag{2.1.5}$$

where $d(n)$ counts the degeneracy of BPS states with charge parameter n. We can retrieve the coefficients $d(n)$ via the Fourier inversion formula

$$d(n) = \int_{\mathcal{C}} e^{-2\pi i n \tau} Z(\tau)\, d\tau \tag{2.1.6}$$

where $\mathcal{C}$ is a contour in $\mathbb{H}$ given by $0 \leq \mathrm{Re}(\tau) < 1$, with fixed imaginary part. Since $Z(\tau)$ is holomorphic on $\mathbb{H}$ except at the cusp, the contour can be deformed freely without changing the value of the integral. The verification of this, as well as the physical interpretation of the $d(n)$ coefficients can be seen in Appendix C.3.

2.2 The Combinatorics

2.2.1 Partitions of Integers

Recall that a **partition** of a positive integer N is a way of writing N as an ordered (non-increasing) sum of positive integers:

$$N = \lambda_1 + \lambda_2 + \cdots + \lambda_k, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1 \tag{2.2.1}$$

The number of such partitions is $p(N)$. The generating function is:

$$\sum_{N=0}^{\infty} p(N)\, q^N = \prod_{n=1}^{\infty} \frac{1}{1-q^n} \tag{2.2.2}$$

2.2.2 Colored Partitions

A **colored partition** of N using r colors is a partition where each integer comes in r distinguishable colors. Equivalently, it is a partition of N using integers from the alphabet $\{1_1, 1_2, \ldots, 1_r, 2_1, \ldots, 2_r, \ldots\}$ where the subscript denotes the color. The number of such partitions is $p_r(N)$, with generating function:

$$\sum_{N=0}^{\infty} p_r(N)\, q^N = \prod_{n=1}^{\infty} \frac{1}{(1-q^n)^r} \tag{2.2.3}$$

2.2.3 Identification $d(n) = p_{24}(n+1)$

Comparing with our partition function:

$$Z(\tau) = q^{-1} \prod_{n=1}^{\infty} \frac{1}{(1-q^n)^{24}} = q^{-1} \sum_{N=0}^{\infty} p_{24}(N)\, q^N = \sum_{N=0}^{\infty} p_{24}(N)\, q^{N-1} \tag{2.2.4}$$

Setting $n = N - 1$ (so $N = n + 1$):

$$Z(\tau) = \sum_{n=-1}^{\infty} p_{24}(n+1)\, q^n \tag{2.2.5}$$

Comparing with (C.3.1), we read off:

$$\boxed{d(n) = p_{24}(n+1)} \qquad n \geq 0 \tag{2.2.6}$$

The degeneracy of the BPS state with charge invariant n equals the number of ways to partition the integer $n+1$ using integers of 24 colors.

The Boltzmann relation $S = \log \Omega$ tells us that the macroscopic entropy of the black hole is $S(n) = \log d(n)$. Using the asymptotic formula for $p_r(N)$, we can derive the asymptotic growth of $d(n)$, and hence of the entropy $S(n)$. This is done in Appendix C.4.2. We immediately read off the leading asymptotic growth of the entropy:

$$S(n) \sim 4\pi\sqrt{n} \qquad \text{as } n \to \infty \tag{2.2.7}$$

This matches the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = \frac{A}{4G_N}$, which also grows as $\sqrt{n}$ for large n.

2.3 Duality and a Second Counting

2.3.1 The D1-D5 System

The DH states have a dual description in the Type-II frame. In this description, the degeneracy $d(m)$ counts the number of such bound states, which equals the **orbifold Euler characteristic**:

$$d(m) = \chi\big(\mathrm{Sym}^{m+1}(K3)\big) \tag{2.3.1}$$

the Euler characteristic of the symmetric product of $(m+1)$ copies of the $K3$ surface. The generating function is:

$$\hat{Z}(\sigma) = \sum_{m=-1}^{\infty} \chi\big(\mathrm{Sym}^{m+1}(K3)\big)\, p^m, \qquad p := e^{2\pi i\sigma} \tag{2.3.2}$$

A celebrated computation shows:

$$\hat{Z}(\sigma) = \frac{1}{\Delta(\sigma)} \tag{2.3.3}$$

2.3.2 Duality Consistency

Equations (2.1.4) and (2.3.3) say that two completely different counting problems- one counting perturbative heterotic string states, the other counting D-brane bound states- give identical partition functions $1/\Delta$. The heterotic string on $T^4 \times T^2$ is dual to Type-II on $K3 \times T^2$, and this duality maps the DH states on one side to the D1-D5 bound states on the other. The equality of the two partition functions was historically one of the first precision checks of heterotic-Type-II duality.

2.4 Jacobi Forms and Elliptic Genera: A brief digression

In the context of two-dimensional superconformal field theories (SCFTs), the elliptic genus is an invariant that captures information about the spectrum of BPS states. For a SCFT with target space M, the elliptic genus is defined as a trace over the Hilbert space of the theory [DMZ12]:

$$\phi_M(\tau, z) = \mathrm{Tr}_{\mathrm{RR}}\left((-1)^F q^H y^J\right) \tag{2.4.1}$$

where $q = e^{2\pi i\tau}$, $y = e^{2\pi i z}$, and F is the fermion number operator. The elliptic genus is a weak Jacobi form of weight 0 and index m, where m depends on the geometry of M. For example, for a K3 surface, the elliptic genus is given by: [DMZ12, Eq. 4.50]

$$\phi_{K3}(\tau, z) = 8\left(\frac{\theta_2(\tau,z)^2}{\theta_2(\tau,0)^2} + \frac{\theta_3(\tau,z)^2}{\theta_3(\tau,0)^2} + \frac{\theta_4(\tau,z)^2}{\theta_4(\tau,0)^2}\right) \tag{2.4.2}$$

where θ_i are the Jacobi theta functions. The Fourier coefficients of ϕ_{K3} encode the degeneracies of BPS states in the SCFT, and their growth can be analyzed using the theory of Jacobi forms. This is just one example of how modular and Jacobi forms arise naturally in the study of quantum black holes and their associated counting problems.

Appendices

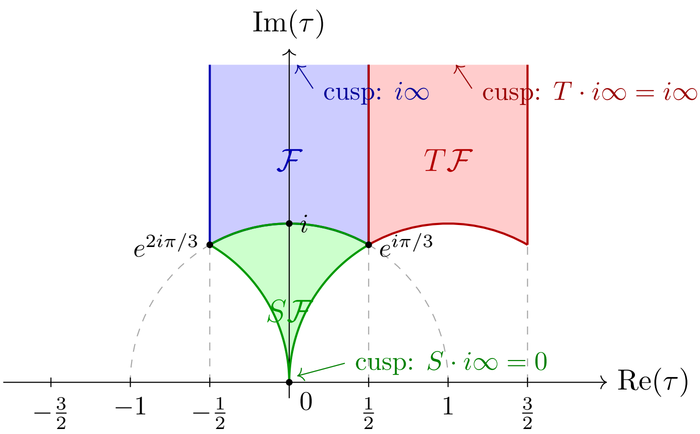
A. Modular Curves and Cusps

A.1 The Problem of Non-Compactness

Modular forms for $SL_2(\mathbb{Z})$ live naturally on the upper half-plane $\mathbb{H}$, where the fundamental domain $\mathcal{F}$ captures the geometry of the quotient $SL_2(\mathbb{Z})\backslash\mathbb{H}$. However, $\mathcal{F}$ is not compact; it possesses a non-compact "end" as $\text{Im}(z) \to \infty$. This non-compactness prevents the direct application of standard results from the theory of compact Riemann surfaces, such as the Riemann-Roch theorem.

Furthermore, the q-expansion $f(z) = \sum_{n=0}^{\infty} a_n q^n$ of a modular form has a "missing point" at $q = 0$, which corresponds to $z = i\infty$. The boundedness condition in the definition of a modular form is essentially a requirement on the behavior at this missing point. When generalizing to congruence subgroups $\Gamma \leq SL_2(\mathbb{Z})$, multiple non-compact ends appear, each contributing a missing point known as a *cusp*.

Note that $\mathcal{F} := \{z \in \mathbb{H} : \text{Re}(z) \leq \frac{1}{2} \text{ and } |z| \geq 1\}$ is *a* fundamental domain for $SL_2(\mathbb{Z})$. There could be more, but they are all related by the action of $SL_2(\mathbb{Z})$, so we can choose this one for convenience. Note that, for $\mathcal{F}$, there is only one non-compact end as $\text{Im}(z) \to \infty$. Any *other fundamental domain*, say $\gamma\mathcal{F}$ for $SL_2(\mathbb{Z})$ will have a similar *missing point* at $\gamma \cdot i\infty$. Depending on $\gamma \in SL_2(\mathbb{Z})$, we either have this missing point at $i\infty$ or at some rational point on the real line.



A.2 The Boundary of $\mathbb{H}$ and the Rational Projective Line

Topologically, the boundary of $\mathbb{H}$ in $\mathbb{C}$ is $\mathbb{R} \cup \{i\infty\} \cong S^1$.

Definition A.2.1. (Projective Line over $\mathbb{Q}$)

$$\mathbb{P}^1(\mathbb{Q}) := \mathbb{Q} \cup \{i\infty\}$$

We represent a rational p/q in lowest terms as $[p : q] \in \mathbb{P}^1(\mathbb{Q})$, with $i\infty$ as $[1 : 0]$. The action of $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ is given by $\gamma \cdot [p : q] = [ap + bq : cp + dq]$.

Lemma A.2.2. Transitivity of the Action $SL_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$.

Proof. For any $[p : q] \in \mathbb{P}^1(\mathbb{Q})$ with $\gcd(p, q) = 1$, we seek $\gamma \in SL_2(\mathbb{Z})$ such that $\gamma \cdot (p/q) = i\infty$, implying the denominator $cp + dq = 0$. Setting $c = q$ and $d = -p$, we have $\gcd(c, d) = 1$. By Bezout's theorem, there exist a, b such that $ad - bc = 1$, thus $\gamma = \begin{pmatrix} a & b \\ q & -p \end{pmatrix} \in SL_2(\mathbb{Z})$ sends p/q to $i\infty$. $\square$

Corollary A.2.3. $SL_2(\mathbb{Z})$ has exactly **one** cusp, as $\mathbb{P}^1(\mathbb{Q})$ is a single orbit that collapses to one point in the quotient.

Note that, for a congruence subgroup Γ, the fundamental domain $F|_\Gamma$ is a region in $\mathbb{H}$ that contains exactly one representative from each orbit of the action of Γ on $\mathbb{H}$. Since $[SL_2(\mathbb{Z}) : \Gamma] < \infty$, we can write $F|_\gamma = \cup_{i=1}^q \gamma_i \mathcal{F}$ (where $\mathcal{F}$ is the standard fundamental domain for $SL_2(\mathbb{Z})$). Each $\gamma_i \mathcal{F}$ has a cusp at $\gamma_i \cdot i\infty$. Thus, the number of cusps of $\Gamma \leq [SL_2(\mathbb{Z}) : \Gamma]$.

Definition A.2.4. (Cusps of Γ) Let $\Gamma \leq SL_2(\mathbb{Z})$ be a subgroup. The *cusps of* Γ are the orbits of the action of Γ on $\mathbb{P}^1(\mathbb{Q})$:

$$\mathrm{Cusps}(\Gamma) := \Gamma \backslash \mathbb{P}^1(\mathbb{Q})$$

For a finite-index subgroup, there are finitely many such cusps.

B. Appendix: Fourier Analysis on the Circle

Lemma B.0.1. Let f be $\mathbb{Z}$-periodic with $|c_n| = O(n^{-2})$ and $|f(x+h) - f(x)| \leq C|h|^e$ for $e > 1/2$. If $c_n = 0$ for all n, then $f \equiv 0$.

Lemma B.0.2 (Parseval)**.**

$$\int_0^1 |f(x)|^2 dx = \sum_{n \in \mathbb{Z}} |c_n|^2.$$

B.0.1 (General Discussion) Periodic Functions of a Real Variable

Let $g : \mathbb{R} \to \mathbb{C}$ be rapidly decaying, and define the $\mathbb{Z}$-periodic function

$$f(x) := \sum_{n \in \mathbb{Z}} g(x+n).$$

Unfolding and Poisson summation

Then

$$\int_{-\infty}^{\infty} g(x)\, dx = \sum_{n \in \mathbb{Z}} \int_0^1 g(x+n)\, dx = \int_0^1 f(x)\, dx.$$

The Fourier coefficients of f are

$$f_m = \int_0^1 f(x) e^{-2\pi i m x}\, dx = \sum_{n \in \mathbb{Z}} \int_0^1 g(x+n) e^{-2\pi i m x}\, dx.$$

Changing variables $u = x + n$ gives

$$f_m = \int_{-\infty}^{\infty} g(u) e^{-2\pi i m u}\, du = \hat{g}(m).$$

Hence

$$f(x) = \sum_{n \in \mathbb{Z}} g(x+n) = \sum_{m \in \mathbb{Z}} \hat{g}(m) e^{2\pi i m x}.$$

Setting $x = 0$ yields the Poisson summation formula:

Theorem B.0.3 (Poisson summation)**.**

$$\sum_{n \in \mathbb{Z}} g(n) = \sum_{m \in \mathbb{Z}} \hat{g}(m).$$

Theorem B.0.4. (Fourier basis for $L^2(S^1)$) $e^{2\pi inx}$ form an orthonormal basis of $L^2(S^1)$, and

$$\sum_{m\in\mathbb{Z}} e^{2\pi im(x-y)} = \sum_{n\in\mathbb{Z}} \delta(x-y+n)$$

in the sense of distributions.

Proof. Let $f(\theta) = \sum_{n\in\mathbb{Z}} \delta(\theta+n)$. Then $f_m = 1$, so

$$f(\theta) = \sum_{m\in\mathbb{Z}} e^{2\pi im\theta},$$

and substitute $\theta = x - y$.

□

C. Quantum Black Holes and Number Theory: Physical Background and Motivation

C.1 Physical Background and Motivation

C.1.1 Black Holes in String Theory: A Brief Overview

BPS black holes: BPS black holes are a special class of black hole solutions in string theory that preserve some fraction of the underlying supersymmetry.

C.1.2 Supersymmetry and BPS States

The theories under consideration are compactifications of string theory to four spacetime dimensions with $\mathcal{N} = 4$ supersymmetry, meaning there are $4 \times 4 = 16$ supercharges. The supersymmetry algebra in four dimensions takes the schematic form:

$$\{Q^I_\alpha, \bar{Q}_{\dot{\alpha}J}\} = \delta^I_J \sigma^\mu_{\alpha\dot{\alpha}} P_\mu \qquad I, J = 1, \ldots, 4 \tag{C.1.1}$$

A BPS state is one for which some of the supercharges Q^I_α annihilate it. The fraction of supercharges preserved determines the "BPS-ness":

- A **half-BPS state** preserves 1/2 of the supercharges (i.e. 8 out of 16). Such states are the simplest and most constrained.
- A **quarter-BPS state** preserves 1/4 of the supercharges (i.e. 4 out of 16). It carries both electric and magnetic charges that are not parallel, and is strictly more complex. Its mass satisfies $M = \max(|Z_1|, |Z_2|)$ where Z_1, Z_2 are the two central charges.

C.1.3 The Compactification: Type-II on $K3 \times T^2$

The specific theory we study is Type-II superstring theory compactified on $K3 \times T^2$, where $K3$ is the unique (up to diffeomorphism) compact complex surface with trivial canonical bundle and $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is a two-torus. This compactification preserves $\mathcal{N} = 4$ supersymmetry in four dimensions. A BPS state in this theory is characterised by its charge vector $(\vec{Q}, \vec{P})$ where $\vec{Q}$ is the electric charge and $\vec{P}$ the magnetic charge, both living in an even self-dual lattice $\Gamma^{6,22}$ [DMZ12, Section 2]. The relevant T-duality invariants are:

$$n = \frac{\vec{Q}^2}{2}, \qquad m = \frac{\vec{P}^2}{2}, \qquad \ell = \vec{Q}\cdot\vec{P} \tag{C.1.2}$$

The half-BPS states we consider in this chapter have $m = 0$ (zero magnetic charge) but $n > 0$, and are purely electric. They are the **Dabholkar–Harvey (DH) states**.

C.1.4 Partition Function

The key physical mechanism linking black hole state counting to modular forms is the following. The degeneracy of DH states is computed by a trace over the Hilbert space of a *chiral conformal field theory* living on the string worldsheet. The partition function of the worldsheet CFT is:

$$Z(\tau) = Tr_{\mathcal{H}}\left(q^{H}\right), \qquad q := e^{2\pi i\tau} \tag{C.1.3}$$

where H is the worldsheet Hamiltonian and the trace is over the physical Hilbert space. The individual degeneracies are then Fourier coefficients:

$$Z(\tau) = \sum_{n} d(n)\, q^{n} \tag{C.1.4}$$

and the black hole entropy is $S(n) = \log d(n)$ for large n.

C.2 The Half-BPS Partition Function $Z(\tau) = 1/\Delta(\tau)$

C.2.1 The Hilbert Space: 24 Free Bosons

We now compute $Z(\tau)$ explicitly. The DH states with $m = 0$ are counted by the partition function of a very specific CFT: the chiral theory of 24 left-moving transverse bosons of the heterotic string.

The Hilbert space $\mathcal{H}$ is the **Fock space** built from bosonic oscillators in 24 transverse spatial directions. For each direction $i \in \{1, \ldots, 24\}$ and each mode number $n \in \{1, 2, 3, \ldots\}$, we have a pair of creation/annihilation operators $a_{in}^{\dagger}$, a_{in} satisfying the canonical commutation relations:

$$[a_{in}, a_{jm}^{\dagger}] = \delta_{ij}\, \delta_{n+m,0} \qquad i, j \in \{1, \ldots, 24\}, \quad n, m \in \{1, 2, \ldots\} \tag{C.2.1}$$

All other commutators vanish: $[a_{in}, a_{jm}] = 0$ and $[a_{in}^{\dagger}, a_{jm}^{\dagger}] = 0$.

The **vacuum state** $|0\rangle$ is defined by:

$$a_{in}|0\rangle = 0 \qquad \forall\, i, n \tag{C.2.2}$$

A general state in $\mathcal{H}$ is obtained by acting on $|0\rangle$ with creation operators:

$$\prod_{i=1}^{24} \prod_{n=1}^{\infty} (a_{in}^{\dagger})^{N_{in}} |0\rangle \tag{C.2.3}$$

where $N_{in} \in \{0, 1, 2, \ldots\}$ are the occupation numbers. Since the oscillators are bosonic, any non-negative integer occupation is allowed.

C.2.2 Computing the Partition Function

The partition function is:

$$Z(\tau) = Tr_{\mathcal{H}}\left(q^{H}\right) = Tr_{\mathcal{H}}\left(\exp(2\pi i\tau H)\right) \tag{C.2.4}$$

We compute this trace mode by mode, using the fact that the modes are independent (the Hilbert space is a tensor product over all (i, n)).

Contribution of a single mode (i, n)**:** The states of mode (i, n) alone are $\{(a_{in}^{\dagger})^{N}|0_{in}\rangle : N = 0, 1, 2, \ldots\}$ with energies $\{nN : N = 0, 1, 2, \ldots\}$. The trace over this factor is:

$$Tr_{(i,n)}\left(q^{n\hat{N}_{in}}\right) = \sum_{N=0}^{\infty} q^{nN} = 1 + q^{n} + q^{2n} + q^{3n} + \cdots = \frac{1}{1 - q^{n}} \tag{C.2.5}$$

This is a geometric series, convergent for $|q| < 1$, i.e. for $\mathrm{Im}(\tau) > 0$, which is exactly the condition that $\tau \in \mathbb{H}$.

The full trace: Including the zero-point energy q^{-1} and multiplying over all 24 directions and all mode numbers $n \geq 1$:

$$Z(\tau) = q^{-1} \prod_{i=1}^{24} \prod_{n=1}^{\infty} \frac{1}{1-q^n} = q^{-1} \prod_{n=1}^{\infty} \frac{1}{(1-q^n)^{24}} \tag{C.2.6}$$

(The product over i just raises $(1-q^n)^{-1}$ to the 24th power, since all 24 directions contribute identically.) Recall the discriminant function:

$$\Delta(\tau) = q \prod_{n=1}^{\infty} (1-q^n)^{24} \tag{C.2.7}$$

Therefore:

$$q^{-1} \prod_{n=1}^{\infty} \frac{1}{(1-q^n)^{24}} = \frac{1}{q \prod_{n=1}^{\infty} (1-q^n)^{24}} = \frac{1}{\Delta(\tau)} \tag{C.2.8}$$

and we conclude:

$$\boxed{Z(\tau) = \frac{1}{\Delta(\tau)}} \tag{C.2.9}$$

C.2.3 Modular Properties of $Z(\tau)$

We now determine what kind of modular object $Z(\tau)$ is.

Transformation Law

Since $\Delta \in S_{12}(SL_2(\mathbb{Z}))$ is a cusp form of weight 12, it satisfies:

$$\Delta\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^{12}\, \Delta(\tau) \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \tag{C.2.10}$$

Therefore $Z(\tau) = \Delta(\tau)^{-1}$ satisfies:

$$Z\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^{-12}\, Z(\tau) \tag{C.2.11}$$

This is exactly the modular transformation law of weight -12.

Holomorphicity and the Pole at $q = 0$

Since $\Delta(\tau)$ is holomorphic on $\mathbb{H}$ and nonzero on all of $\mathbb{H}$, the function $Z(\tau) = 1/\Delta(\tau)$ is also holomorphic on $\mathbb{H}$ with no poles there.

The issue is at the boundary point $\tau = i\infty$. Since Δ is a *cusp form*, $\Delta(\tau) \to 0$ as $\tau \to i\infty$, i.e. as $q \to 0$. More precisely:

$$\Delta(\tau) = q - 24q^2 + 252q^3 - 1472q^4 + \cdots = q(1 + \mathcal{O}(q)) \tag{C.2.12}$$

so Δ has a **simple zero** at $q = 0$. Consequently:

$$Z(\tau) = \frac{1}{\Delta(\tau)} = \frac{1}{q(1+\mathcal{O}(q))} = q^{-1}(1+\mathcal{O}(q)) \tag{C.2.13}$$

$Z(\tau)$ has a **simple pole** at $q = 0$, i.e. at $\tau = i\infty$. It is therefore *not* bounded as $\mathrm{Im}(\tau) \to \infty$, and so $Z(\tau)$ is not a holomorphic modular form in the strict sense.

C.3 The Degeneracy $d(n)$ and its Extraction

C.3.1 Definition of the Degeneracy

The q-expansion of $Z(\tau)$ defines the degeneracies:

$$Z(\tau) = \frac{1}{\Delta(\tau)} = \sum_{n=-1}^{\infty} d(n)\, q^n \tag{C.3.1}$$

The first few values, computed directly from the product formula, are:

$$Z(\tau) = q^{-1} + 24 + 324\,q + 3200\,q^2 + 25650\,q^3 + 176256\,q^4 + \cdots \tag{C.3.2}$$

so $d(-1) = 1$, $d(0) = 24$, $d(1) = 324$, $d(2) = 3200$, and so on.

The physical interpretation: $d(n)$ counts the number of DH states with charge invariant $N^2/2 = n$. The state with $n = -1$ (energy $E = -2$) is unphysical; the state with $n = 0$ corresponds to the massless states, and for $n \geq 1$ these are massive BPS states.

C.3.2 The Contour Integral

The degeneracy $d(n)$ is recovered from $Z(\tau)$ by the contour integral:

$$d(n) = \int_{\mathcal{C}} e^{-2\pi i\tau n}\, Z(\tau)\, d\tau \tag{C.3.3}$$

where the contour $\mathcal{C}$ is a horizontal line in $\mathbb{H}$:

$$\mathcal{C} = \{\tau \in \mathbb{H} : 0 \leq \mathrm{Re}(\tau) < 1, \quad \mathrm{Im}(\tau) = \epsilon\} \tag{C.3.4}$$

for some fixed $\epsilon > 0$.

Let us verify this. Write $Z(\tau) = \sum_{m=-1}^{\infty} d(m) q^m$ and $q = e^{2\pi i\tau}$. Then:

$$\begin{aligned}
\int_{\mathcal{C}} e^{-2\pi i\tau n} Z(\tau)\, d\tau &= \int_0^1 e^{-2\pi i(\sigma+i\epsilon)n} \left(\sum_{m=-1}^{\infty} d(m) e^{2\pi i(\sigma+i\epsilon)m} \right) d\sigma \\
&= \sum_{m=-1}^{\infty} d(m)\, e^{-2\pi\epsilon(n-m)} \int_0^1 e^{2\pi i\sigma(m-n)}\, d\sigma \\
&= \sum_{m=-1}^{\infty} d(m)\, e^{-2\pi\epsilon(n-m)}\, \delta_{mn} \\
&= d(n)
\end{aligned} \tag{C.3.5}$$

where we used $\int_0^1 e^{2\pi i\sigma(m-n)} d\sigma = \delta_{mn}$ and the exponential factor evaluates to 1 when $m = n$.

C.4 The Entropy $S(n)$ and the Hardy–Ramanujan Formula

C.4.1 The Boltzmann Relation

For large charge invariant n, the DH states correspond to the microstates of a **small black hole** in the supergravity description. The macroscopic entropy is given by the Bekenstein–Hawking formula, and the microscopic entropy is:

$$S(n) = \log d(n) \qquad n \gg 1 \tag{C.4.1}$$

A precise match between macroscopic and microscopic entropy is one of the key tests of string theory as a theory of quantum gravity.

C.4.2 The Hardy–Ramanujan Asymptotic Formula

The asymptotic growth of $d(n) = p_{24}(n+1)$ can be computed by the **Hardy–Ramanujan circle method** (also called the Cardy formula in the CFT context). The method uses the modular properties of $Z(\tau)$ in a crucial way.

The Modular Transformation

The starting point is the transformation of $Z(\tau)$ under $S : \tau \mapsto -1/\tau$:

$$Z(-1/\tau) = (-1/\tau)^{-12} \cdot \frac{1}{\Delta(-1/\tau)} = \tau^{12} \cdot \frac{1}{\tau^{-12}\Delta(\tau)} = \tau^{-12} \cdot \frac{\tau^{12}}{\Delta(\tau)} \cdot \tau^{12} \tag{C.4.2}$$

More carefully: since $\Delta(-1/\tau) = \tau^{12}\Delta(\tau)$:

$$Z(-1/\tau) = \frac{1}{\Delta(-1/\tau)} = \frac{1}{\tau^{12}\Delta(\tau)} = \tau^{-12}\, Z(\tau) \tag{C.4.3}$$

This transformation relates the behavior of $Z(\tau)$ near $\tau = i\infty$ (i.e. $q \to 0$, $\mathrm{Im}(\tau) \to \infty$) to its behavior near $\tau = 0$ (i.e. $-1/\tau \to i\infty$). The key point is that the pole of $Z(\tau)$ at $q = 0$ maps, under S, to a pole of $Z(-1/\tau)$ at $\tau = 0$.

Saddle Point Approximation

The contour integral for $d(n)$ is:

$$d(n) = \int_0^1 e^{-2\pi i\tau n}\, Z(\tau)\, d\tau \bigg|_{\mathrm{Im}(\tau)=\epsilon} \tag{C.4.4}$$

For large n, the dominant contribution comes from the region near $\tau = 0$ (the saddle point). Using the transformation (C.4.2) to rewrite $Z(\tau)$ near $\tau = 0$, and performing the integral by the method of steepest descent, one finds:

$$d(n) \sim \exp\bigl(4\pi\sqrt{n}\bigr) \qquad n \to \infty \tag{C.4.5}$$

(The leading exponential; there are subleading power corrections.) More precisely, the Hardy–Ramanujan–Rademacher expansion gives an *exact* formula for $d(n)$, not just an asymptotic one, but the leading term is:

$$d(n) \sim \frac{1}{2} \cdot \frac{e^{4\pi\sqrt{n}}}{n^{27/4}} \qquad n \to \infty \tag{C.4.6}$$

including the polynomial prefactor.

The Entropy

From (C.4.6) and the Boltzmann relation (C.4.1):

$$S(n) = \log d(n) \sim 4\pi\sqrt{n} \qquad n \gg 1 \tag{C.4.7}$$

ignoring sub-leading terms.

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